Possible design solutions

X-ray

There is a slot in both the bottom and both sides for the X-ray plate to go into. The plate measures 15.75” x 12” and is around an inch thick.

- One solution is to have the plates made out of polypropylene for both its strength and low density to not show up on X-rays.
- For the X-ray from the top, the keepers will set the box on top of the plate and take the X-ray. One solution for X-rays from the side is to have a track toward the bottom of the sides to hold the X-ray plates.

X-Raying through the box

Through talking with the zookeepers, vet, and other sources, we found a plastic-based box with an aluminum frame on the outside. One point that the Komodo dragon keepers brought up was the ability to easily clean out the boxes between uses. This caused the need for a plastic material rather than wood. The ability to X-ray through the box causes the aluminum frame to only be on the outside edges to not interfere with the X-ray. This causes an issue with the doors as we can not have any metal in the middle of the walls. This disallows

Ultrasound & Drawing Blood

There needs to be a door 6” x 18” for the keepers to be able to reach in and perform an ultrasound on the dragons

- Have the door slide up out of the way. This would require the door to have some handle at the top of the box to slide the door up, it would also need some way of keeping the door from coming back down. It would also require a lock to keep the door from accidentally opening.
- Have a spit door that slides to the sides, this would require a slot for the doors to run in along the sides of the box and would need a way of securing it shut to keep the door from accidentally opening.

Design Solution Choices:

All members must use same criteria and weights and the same design options in the below weighted decision matrices

vote 1-10 for each matrix

Sam	**Weights**	**Horizontal Sliding Door**	**Vertical Sliding Door**	**Design Hinge-Latch Door**	**Divergent Sliding Door**
Simplicity of latching/opening	0.15	1	0	0.5	0
Latching ability	0.15	1	1	1	0.5
X-Ray Compatibility	0.25	1	1	0.5	0.5
Range of operator motion	0.2	0	1	1	1
Durability	0.15	1	1	0	0
Reliability	0.1	1	1	1	0
	Total Score	0.8	0.85	0.65	0.4

Parsa	**Weights**	**Horizontal Sliding Door**	**Vertical Sliding Door**	**Design Hinge-Latch Door**	**Divergent Sliding Door**
Simplicity of latching/opening	0.15	1	0	0.75	0.5
Latching ability	0.15	0.5	1	0.5	0.2
X-Ray Compatibility	0.25	1	1	0.5	1
Range of operator motion	0.2	0.5	1	1	1
Durability	0.15	0.8	1	0	0.2
Reliability	0.1	1	1	0.25	0
	Total Score	0.795	0.85	0.5375	0.585

Shanti	**Weights**	**Horizontal Sliding Door**	**Vertical Sliding Door**	**Design Hinge-Latch Door**	**Divergent Sliding Door**
Simplicity of latching/opening	0.15	1	1	0.4	0.2
Latching ability	0.15	0.5	0.4	0.2	0.2
X-Ray Compatibility	0.25	0.4	0.8	0.2	0.1
Range of operator motion	0.2	0.2	0.4	0.2	0.4
Durability	0.15	0.4	0.5	0.4	0.2
Reliability	0.1	0.6	0.5	0.7	0
	Total Score	0.485	0.615	0.31	0.195

Step 3: check out your team's results!		**Horizontal Sliding Door**	**Vertical Sliding Door**	**Design Hinge-Latch Door**	**Divergent Sliding Door**
	total team scores	2.08	2.315	1.4975	1.18

Using the designs from Element C, team members independently completed a design matrix based on the weighted criteria curated for the zookeepers. One at a time, each team member ranked each design concept by their shared criteria; therefore, every member would have an equal and independent ranking of each idea. As represented in the picture of the data, different team members placed different preferences on different designs, so the design matrices vary greatly in the range of scores and of winning designs. As a result of these differences, we averaged the results of the three matrices resulting in a final design choice of “vertical sliding door”.